Unit – 5 :

Transmission Problems:.....

Introduction to Radio Wave Propagation:

The basic Propagation Mechanisms:

Reflection, Diffraction, Scattering, Path Loss, Shadowing, Time dispersion, Time Alignment, Combined Signal Loss, High Bit Error Rate.

##

Solution To Transmission Problems:.....

Channel Coding:

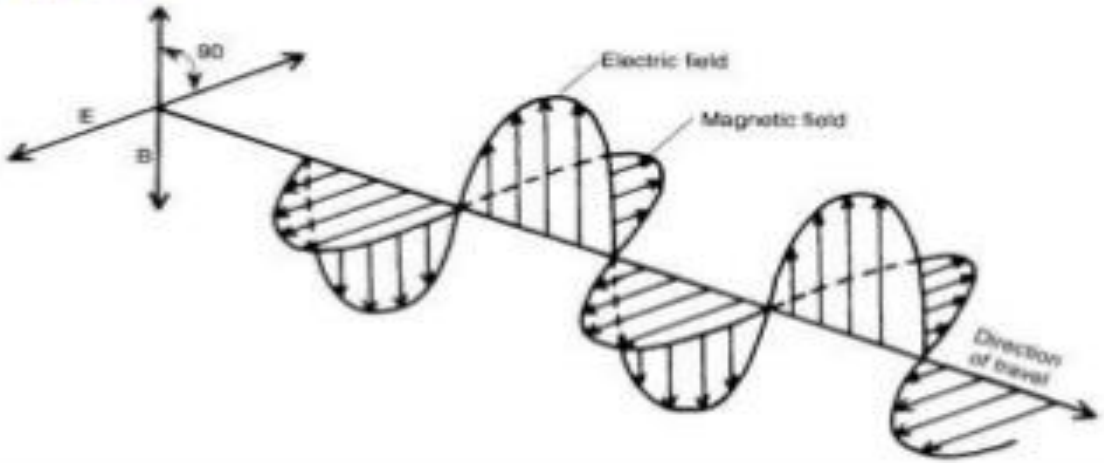
Interleaving:

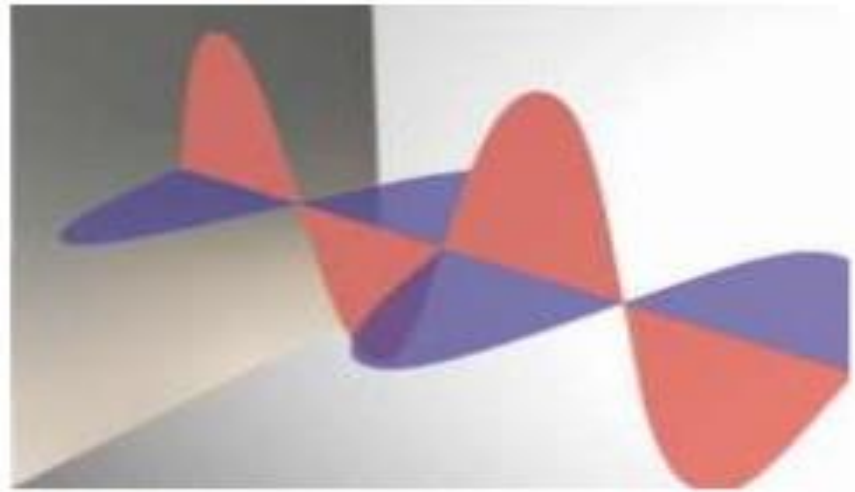
Diversity:

Fundamental of Equalization:

Frequency Hopping:

Radio Wave

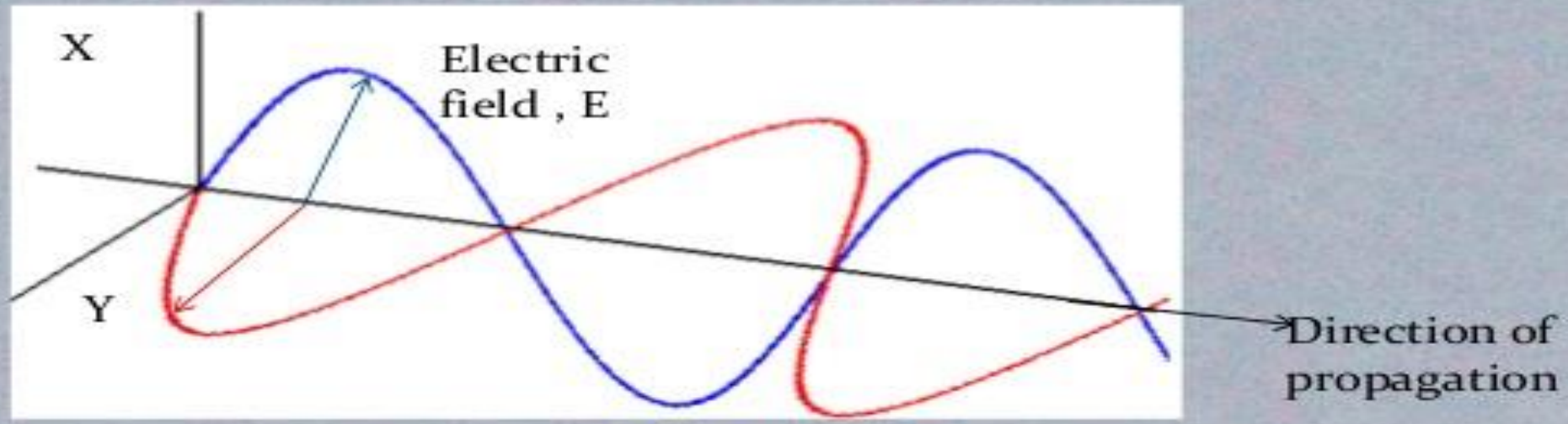




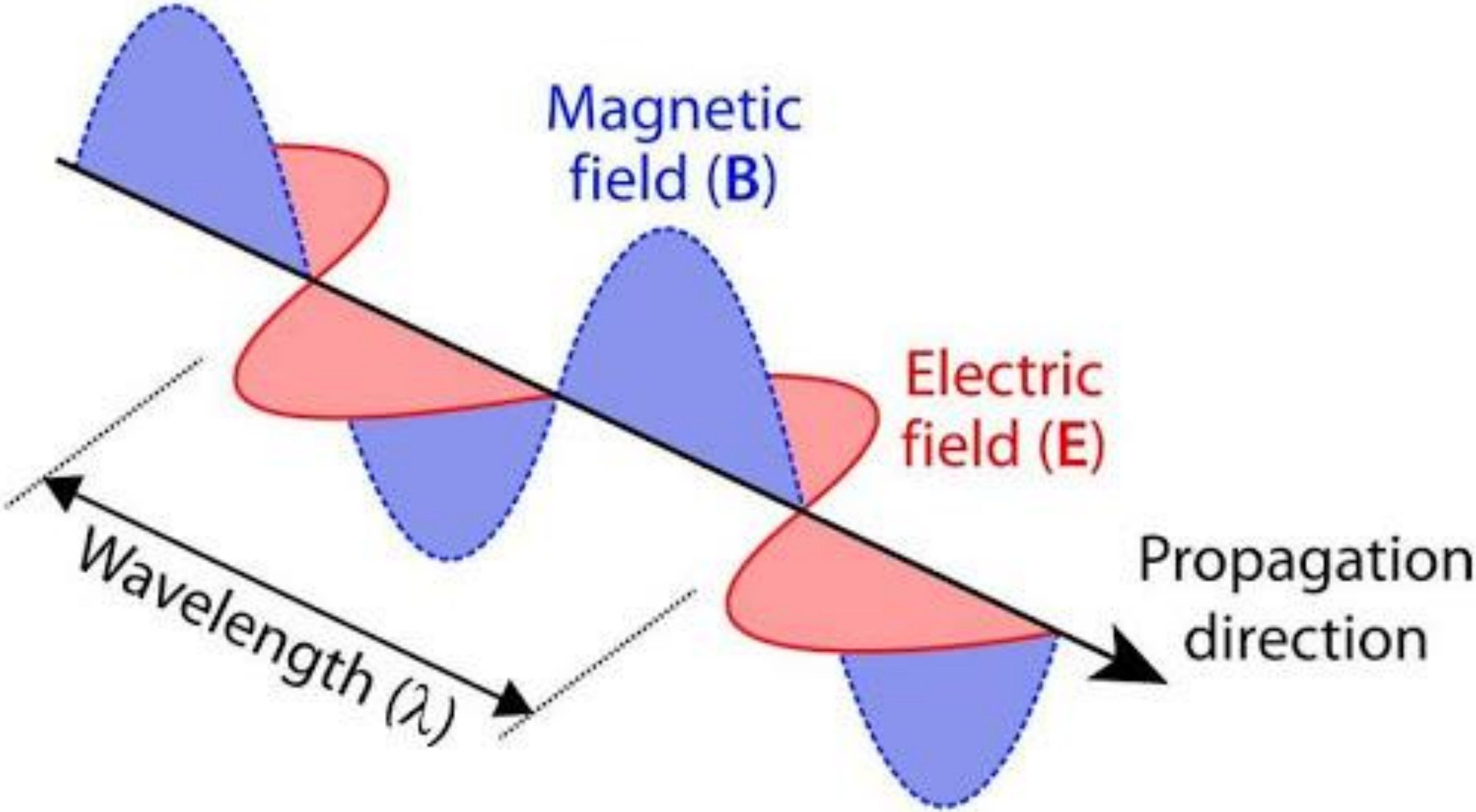
Radio wave is an electromagnetic wave having wavelength between 1mm-30km & having frequency between 10KHz-300 GHz

A radio wave moves outward from the radiator with its electric and magnetic fields at right angles to the direction of the wave-font motion and to each other. These waves moving through free space are transverse electromagnetic (TEM) waves consisting of mutually perpendicular electric and magnetic fields varying and travelling together in synchronism.

INTRODUCTION



- **The term Radio waves arbitrarily applied to electromagnetic waves in the frequency range of 0.001 hertz to 10^{15} hertz and the wavelength ranging from 3×10^{-8} m to 3×10^{11} m.**
- **Radio waves comprises of both Electric and Magnetic fields.**
- **The two fields are at right-angles to each other and the direction of propagation is at right-angles to both fields.**


Magnetic
field (B)
Electric
field (E)
Propagation
direction
Wavelength (λ)

PROPAGATION - INTRO

Propagation: How radio waves travel from point A to point B; and the events occurring in the transmission path that affect the communications between the points, stations, or operators.

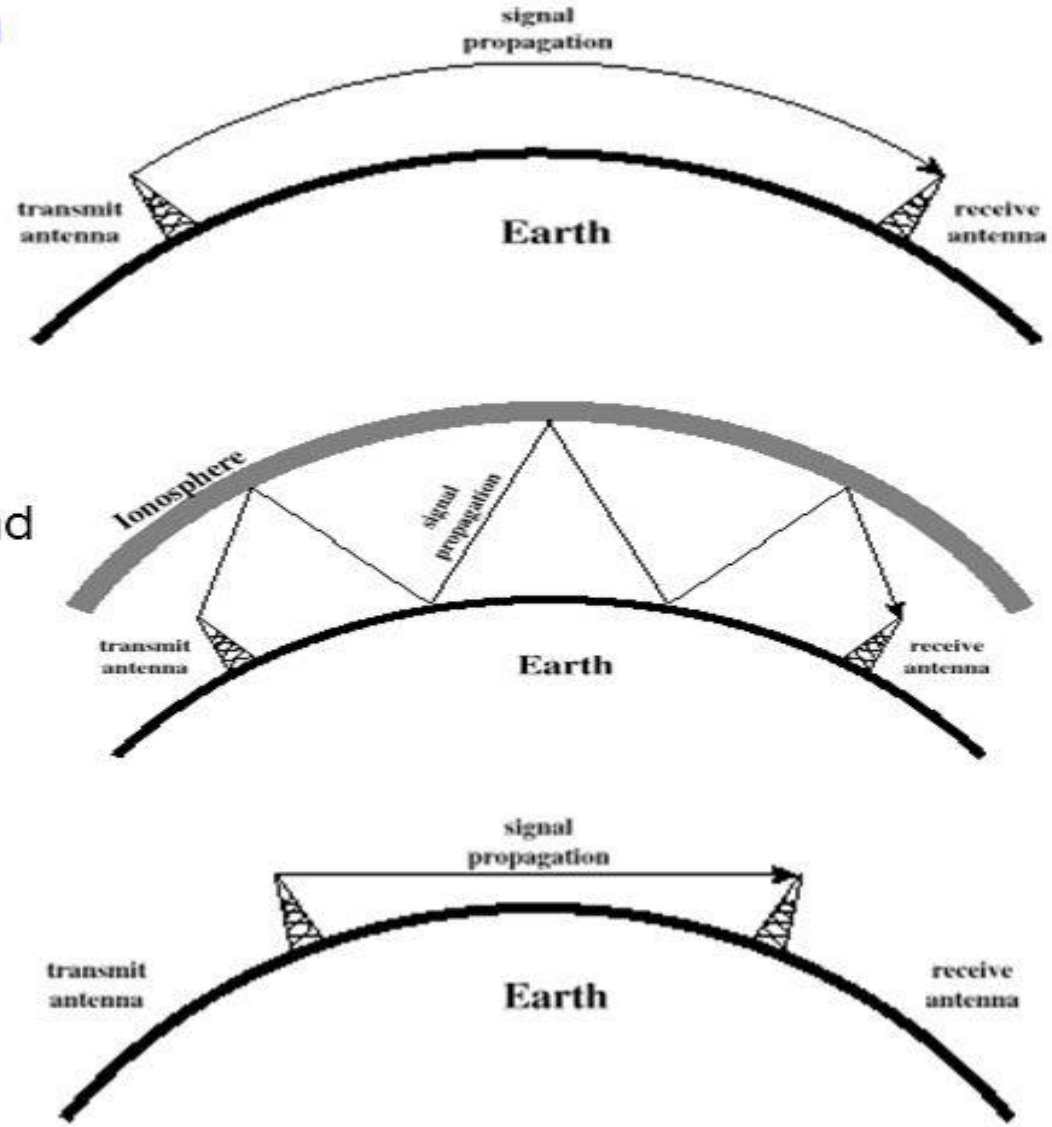
When the electrons in a conductor, (antenna wire) are made to oscillate back and forth, Electromagnetic Waves (EM waves) are produced.

These waves radiate outwards from the source at the speed of light, 300 million meters per second.

Light waves (waves we see) and radio waves (waves we hear)are both EM waves, differing only in frequency and wavelength.

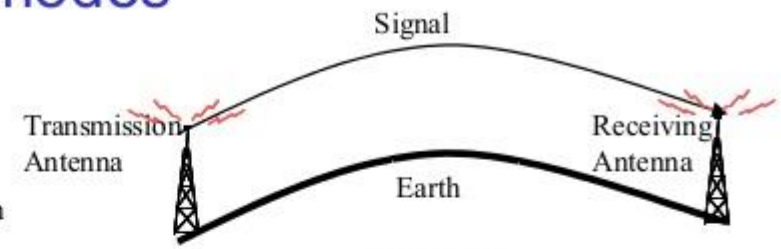
Propagation Modes

- **Radio signal behaves like light in free space**
- **Ground Wave**
  - Frequencies up to 2 MHz
  - Follows contour of the earth
  - Example: AM Radio
- **Sky Wave**
  - Signal reflected from ionosphere and earth's surface
  - Can travel thousands of kilometers
  - Frequency: 2-30MHz
  - Amateur Radio, Military Comm.
- **Line of Sight**
  - Transmitting and receiving antennas must be within line of sight
  - Frequency: More then 30MHz
  - TV, satellite, optical comm.

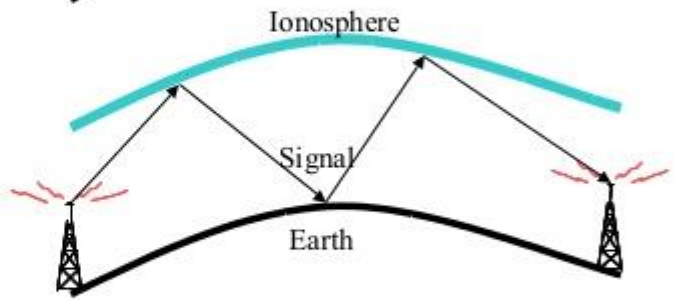


Propagation modes

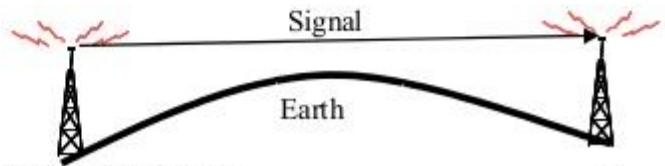
a) Ground Wave Propagation



b) Sky Wave Propagation



c) Line-of-Sight Propagation



Radio propagation definition
Radio propagation is the way **radio waves** travel or **propagate** when they are transmitted from one point to another and affected by the medium in which they travel and in particular the way they **propagate** around the Earth in various parts of the atmosphere.

What are the different types of radio wave propagation?
The mode of propagation of electromagnetic waves in the atmosphere and in free space may be divided into the following three categories:
•The line of sight (LOS) **propagation**.
•Ground **wave propagation**.
•Sky wave **propagation**

What is propagation of radio wave?
Radio propagation is the behavior of **radio waves** as they travel, or are **propagated**, j one point to another, or into various parts of the atmosphere. ... Line-of-sight **propagation** means **radio waves** which travel in a straight line from the transmitting antenna to the receiving antenna.

Why do we need to study radio wave propagation?
The **study** of **radio propagation**, how **radio waves** move in free space and over the surface of the Earth, is vitally important in the design of practical **radio** systems. **Radio waves** passing through different environments experience reflection, refraction, polarization, diffraction, and absorption.

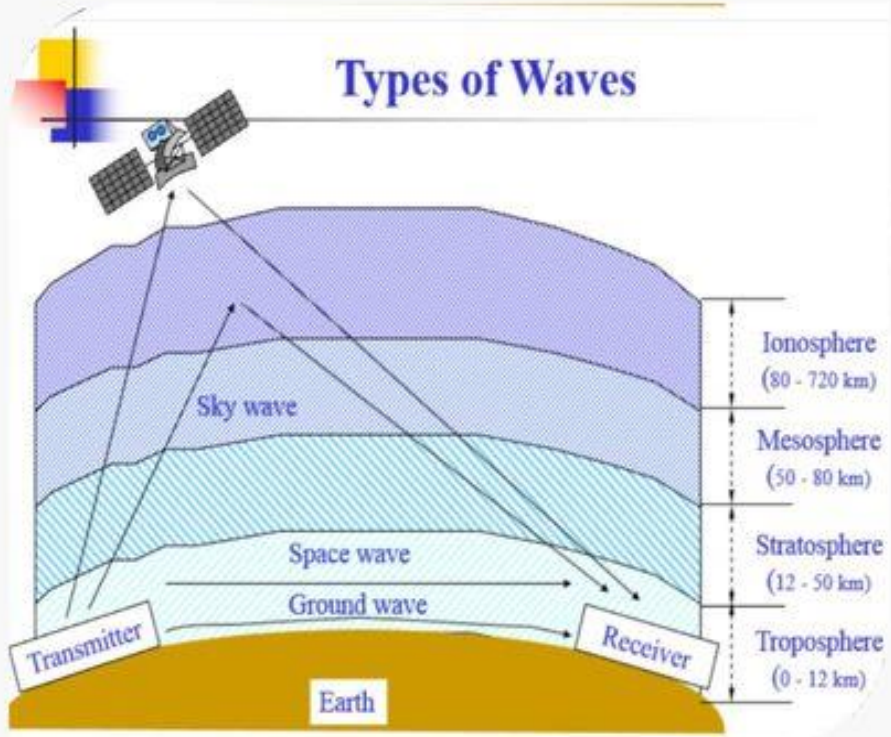
Introduction to Radio Wave Propagation

- The radio waves may undergo reflection, diffraction, and scattering.
- The presence of high-rise buildings causes severe diffraction loss.
- Due to multiple reflections from various objects, the electromagnetic waves travel along different paths of varying lengths causing multipath fading at a specific location.
- The strengths of the waves decrease as the distance between the transmitter and receiver increases.

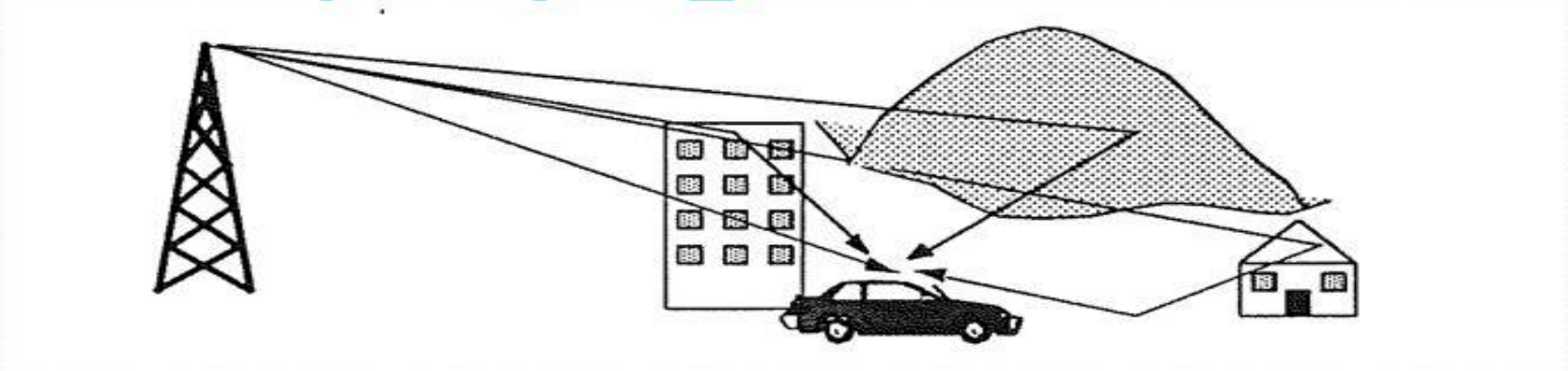
TYPES OF PROPAGATION

Propagation of Radio waves takes place by different modes, the mechanism being different in each case. Based on that, it can be classified as:

1. Ground (Surface) waves
2. Space (Tropospheric) waves
3. Sky (Ionospheric) waves



Radio propagation and fading

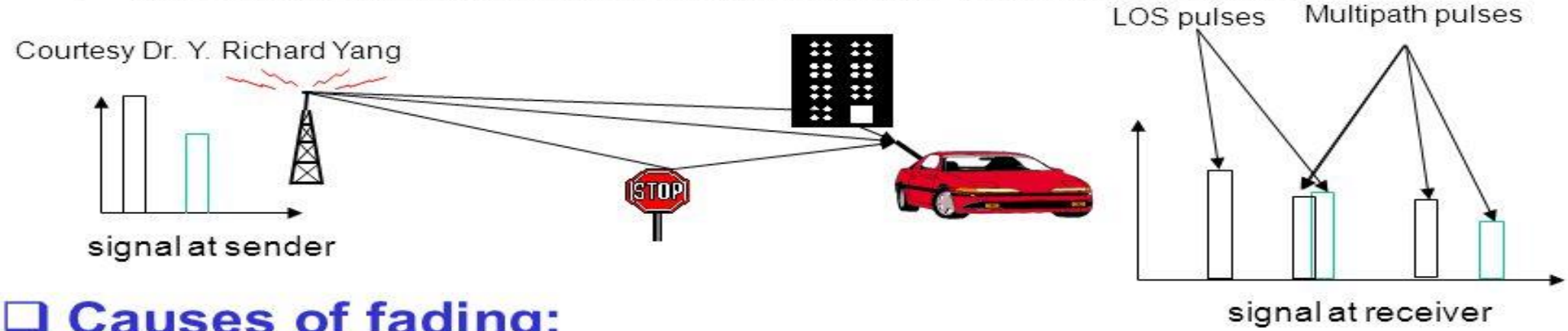


- **Rays coming from different paths**
  - **Reflections**
  - **Refractions**
  - **scattering**
- **Large scale fading**
  - **Path loss**
  - **shadowing**
- **Small scale fading**
  - **Multi-path fading**



What is Fading?

- **The main problem in wireless transmission**
- **Definition:**
  - Time variation of received signal power caused by changes in the transmission medium or path (s).
  - In fixed environment is caused by atmospheric conditions
  - In mobile environments creates more complex effects

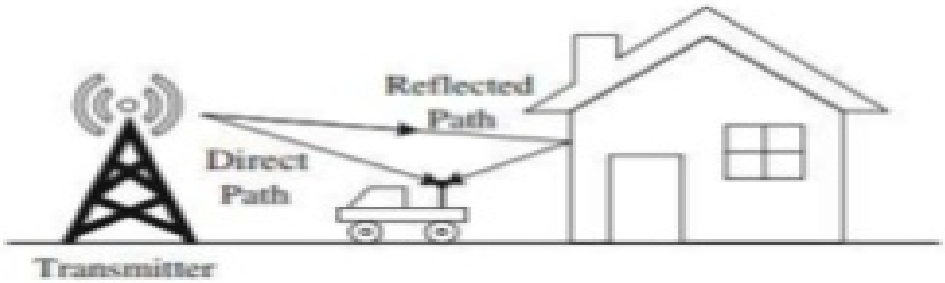


- **Causes of fading:**
  - Free space loss
  - Multi-path propagation
    - ✓ Reflection, scattering, diffraction, refraction
  - Interference with other transmitters
  - Atmospheric absorption
  - Mobility
    - ✓ Fast fading, small fading

Multipath Fading

- Fading is a common phenomenon in wireless communications (especially mobile) , in which the received signal power (suddenly) drops
- Signal fading occurs due to reflection, refraction, scattering, and diffraction
- Multipath fading occurs when a signal reaches the receiver via multiple paths

An example of multipath



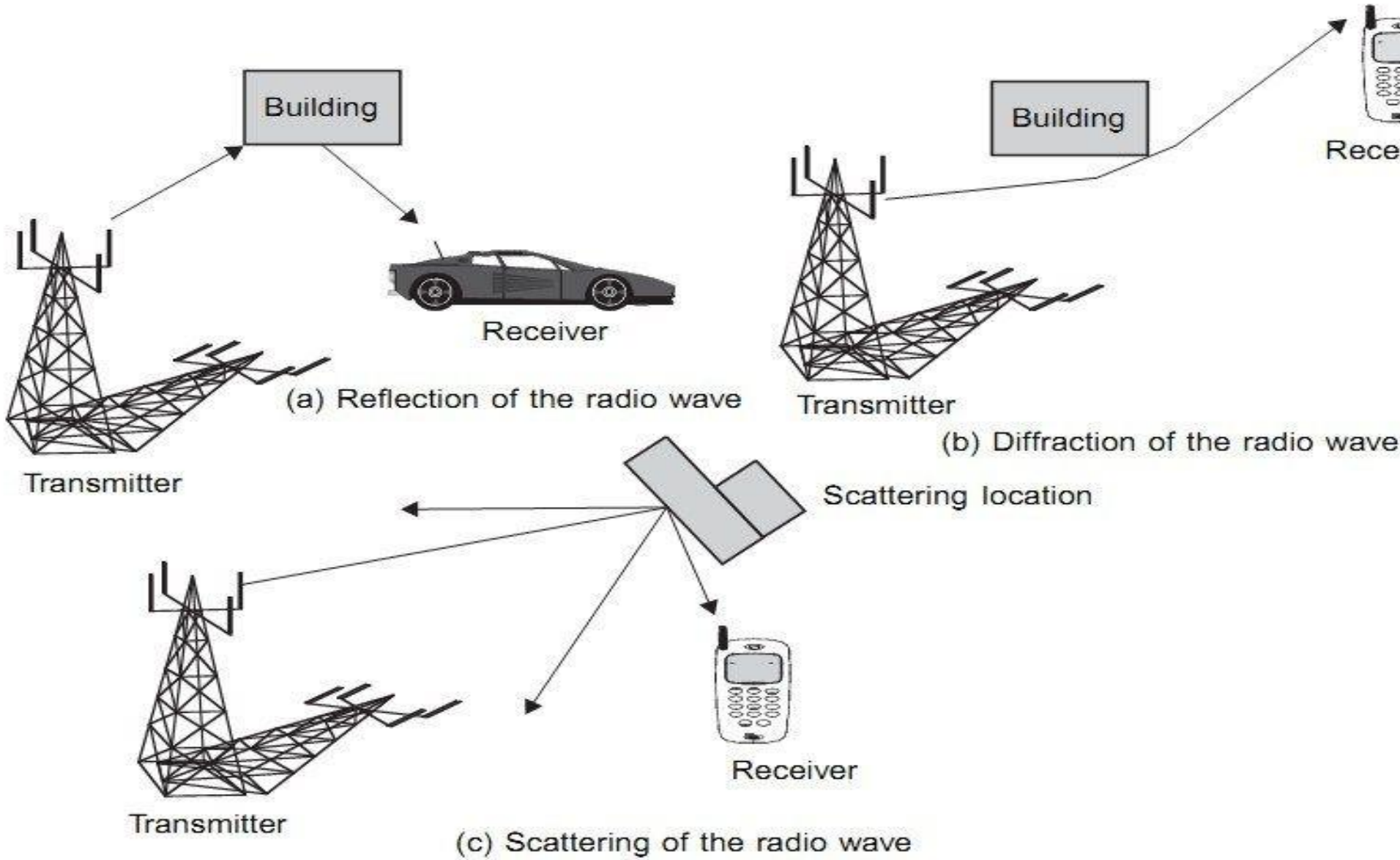
- Because they arrive at different times and phases, the multiple instances of the signal can cancel each other, causing the loss of signal or connection. The problem becomes more severe when higher data rates are explored.

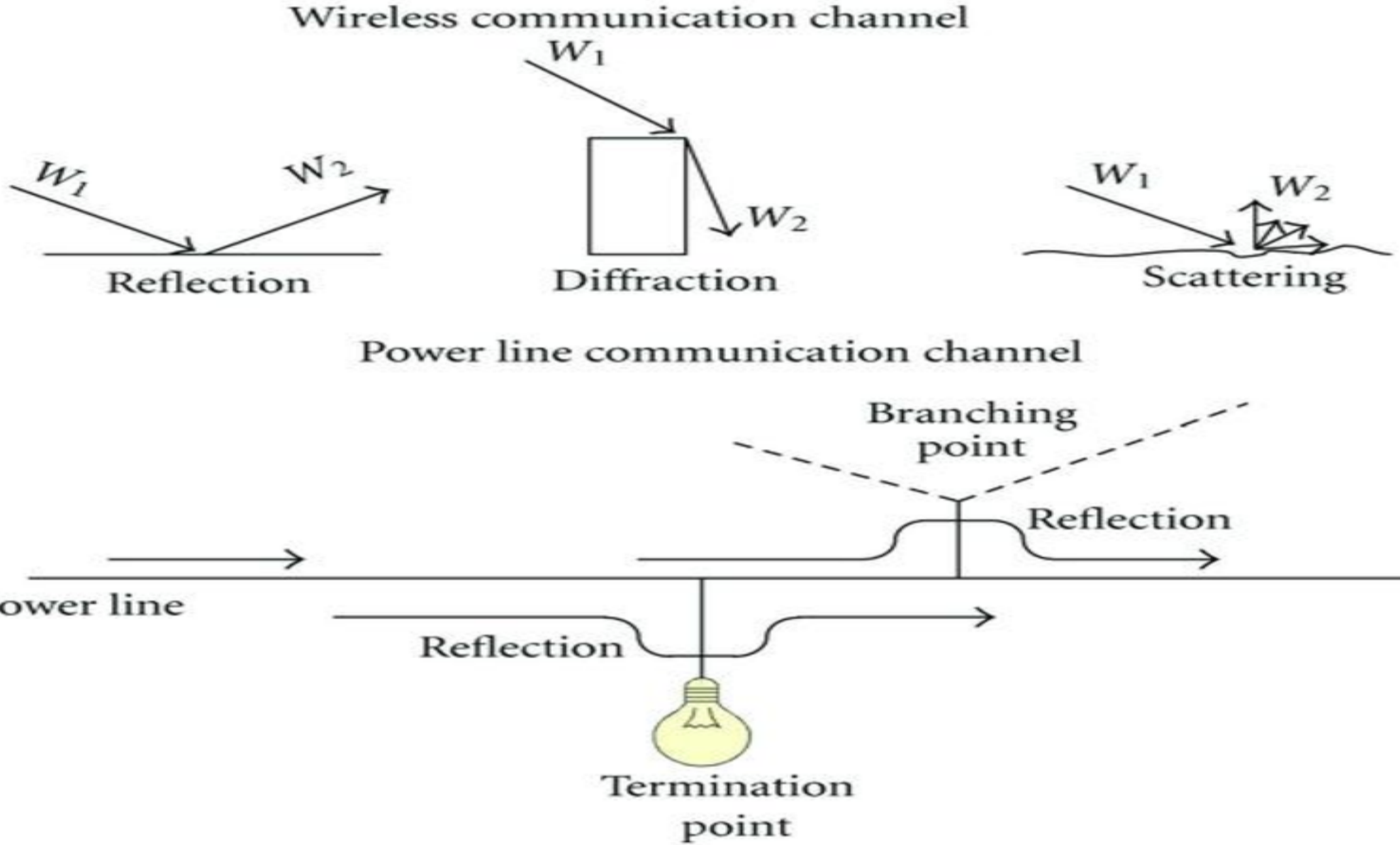
Reflection, **diffraction**, and **scattering** are the three **basic propagation mechanisms** which impact **propagation** in a **mobile communication** system.

Scattered waves are produced by rough surfaces, small objects, or by other irregularities in the channel.

PROPAGATION MECHANISMS

- They have an impact on the wave propagation in a mobile communication system
- The most important parameter, Received power is predicted by large scale propagation models based on the physics of reflection, diffraction and scattering


Building
Receiver
Transmitter
(a) Reflection of the radio wave
Building
Receiver
Transmitter
(b) Diffraction of the radio wave
Scattering location
Receiver
Transmitter
(c) Scattering of the radio wave


Wireless communication channel
W_1
W_1
W_2
Reflection
W_2
Diffraction
W_1
W_2
Scattering
Power line communication channel
Branching point
Reflection
Power line
Reflection
Termination point

THREE BASIC PROPAGATION MECHANISMS

- **Reflection** : occurs when a signal is transmitted, some of the signal power may be reflected back to its origin rather than being carried all the way.
- **Diffraction** :The apparent bending of waves around small obstacles and the spreading out of waves past small openings.
- **Scattering** is a general physical process where light, sound, or moving particles, are forced to deviate from a straight trajectory, by one or more localized non-uniformities, in the medium through which they pass.

- Reflection occurs when a propagating electromagnetic wave impinges upon an object which has a very large dimensions when compared to the wavelength of the propagating wave
- Reflection occur from the surface of the earth and from building and walls

- Diffraction occurs when the radio path between the transmitter and receiver is obstructed by a surface that has a sharp irregularities(edges)
- At high frequency diffraction depends on geometry of the objects as well as amplitude, phase , polarization of the incident wave at the point of diffraction

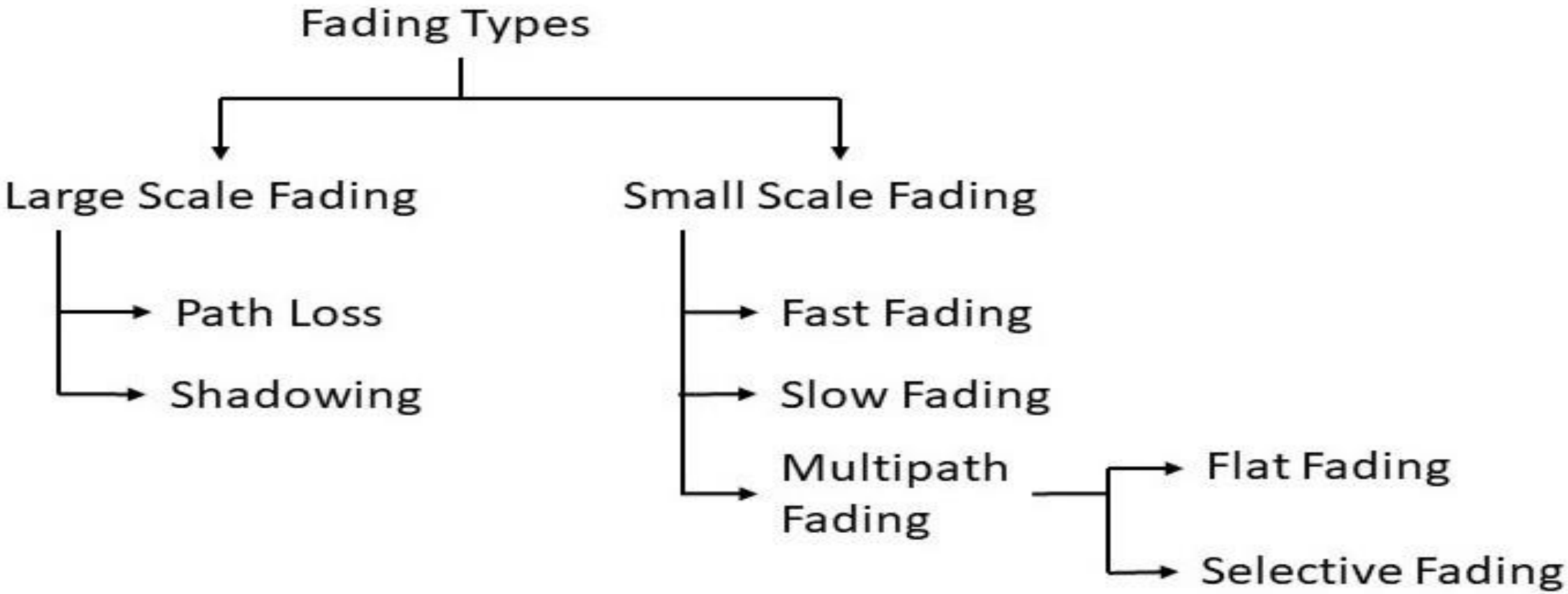
- Scattering occurs when the medium through which the wave travels consists of objects with dimensions that are small compared to the wavelength and where the number of obstacles per unit volume is large
- Scattered waves are produced by rough surfaces, small objects or by other irregularities in the channel

Fading

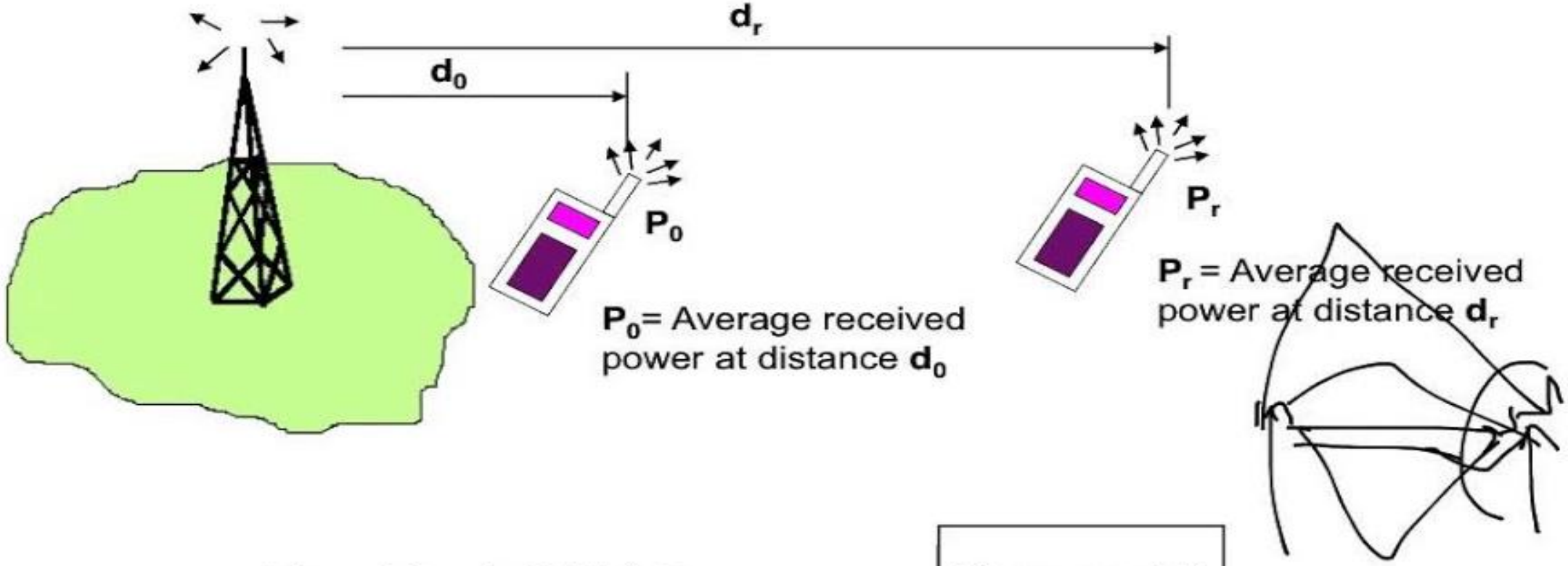
The most common types of fading are

Small scale fading: Small scale fading or simply fading is used to describe the rapid fluctuations of the amplitude, phases or multipath delays of a radio signal over a short period of time or travel distance,

Large scale fading : Large-scale fading is the result of signal attenuation due to signal propagation over large distances and diffraction around large objects in the propagation path.


Fading Types
Large Scale Fading
Small Scale Fading
Path Loss
Shadowing
Fast Fading
Slow Fading
Multipath Fading
Flat Fading
Selective Fading

Simplified Mobile Channel Propagation Model



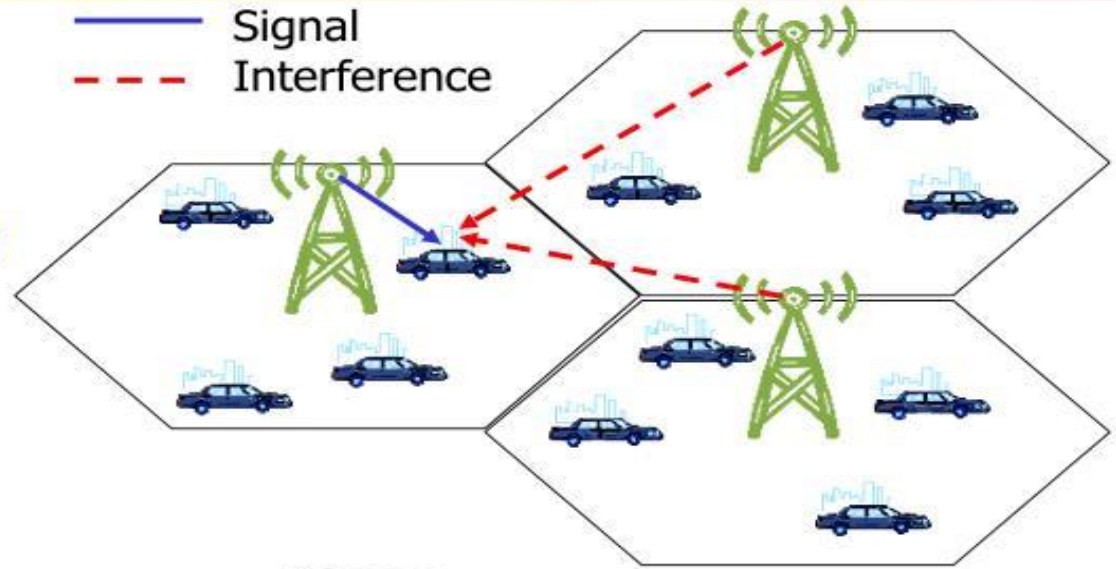
$$P_r = P_0 \ (d_r/d_0)^{-n} \longrightarrow \boxed{P_r = \alpha\, d_r^{-n}}$$

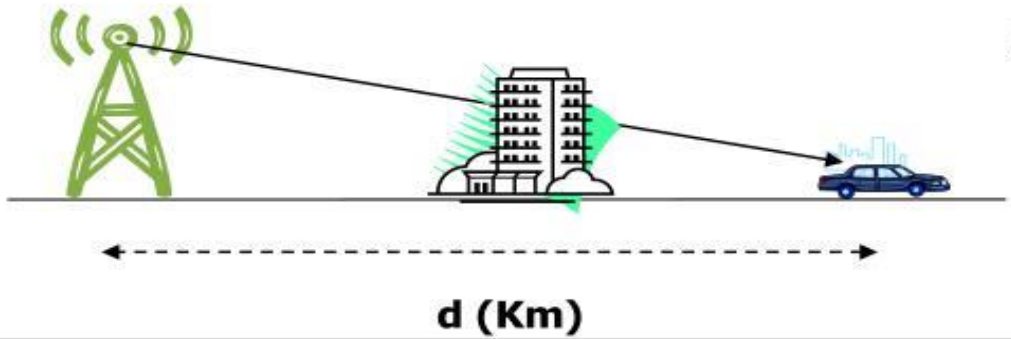
Received signal power P_r is proportional to d_r^{-n}, where **n** is the ***path loss exponent***
In urban cellular systems n ranges from 2 to 4

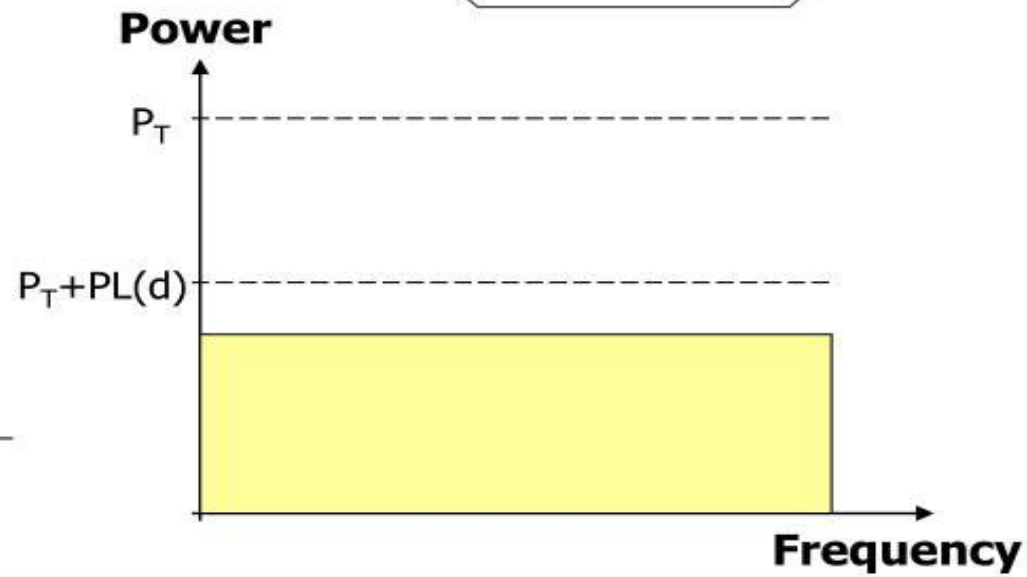
Wireless Communication Channels

Large-Scale Parameters

- Distance Pathloss
- Lognormal Shadowing







What is delay spread in wireless communication?

Delay spread is a measure of the multipath profile of a **mobile communications** channel. It is generally defined as the difference between the time of arrival of the earliest component
(e.g., the line-of-sight wave if there exists)
and
the time of arrival of the latest multipath component.

Multipath and Fading

- The wireless channel is a **multipath** propagation channel.
- Multipath → Rapid Fluctuation of signal amplitude → **SMALL SCALE FADING**
- Caused by destructive interference of two or more versions of the transmitted signal arriving at the receiver at slightly different times (with different amplitudes and phases).
- Signals may come from
  - From different directions
  - With different ***propagation delays*** (random)
  - With different ***amplitudes*** (random)
  - With **phases** (random)
  - With different ***angles of arrival***
- These multipath combine **vectorially → Fade or Distort**

"RANDOM"
"Static"

FADING

Fading : when the Power or Strength of received signal varies with respect to the time.

that is, the power 'fades' (weakens) gradually, due to many reasons like -

Refraction, Multipath propagation, obstacles, attenuation, changes in the transmission medium etc.

It can be -

Long Term or Large Scale Fading

&

Short Term or Small Scale Fading.

The **Bit Error-Rate** Test

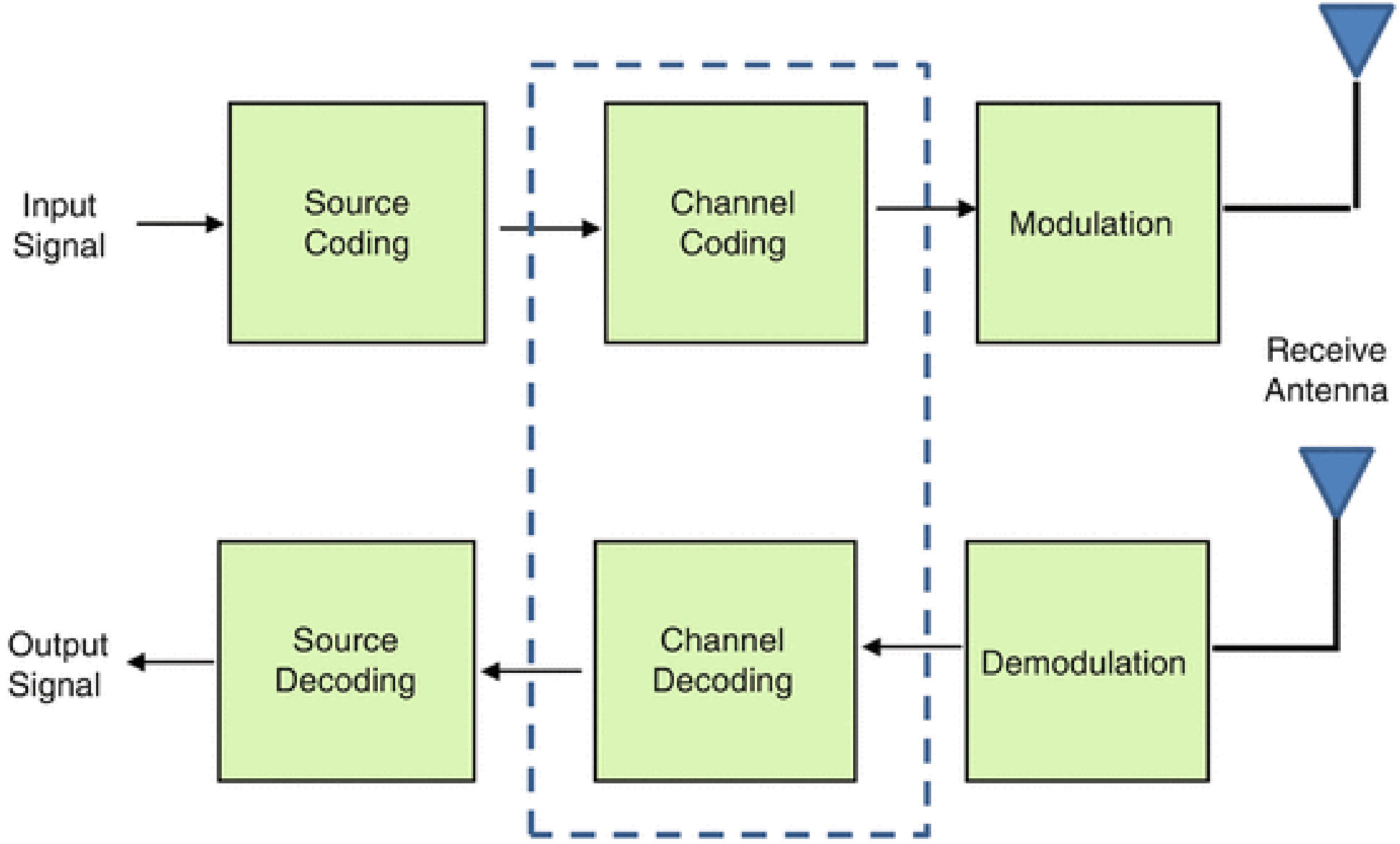
BER is usually expressed as 10 to a negative power. For example, if a **transmission** has a **BER** of 10 to the minus 4, this means that of 10,000 **bits** transmitted, 1 had an **error**. A **high BER** value indicates a noisy line, which can cause poor **network** performance.

Solution To Transmission Problems:.....

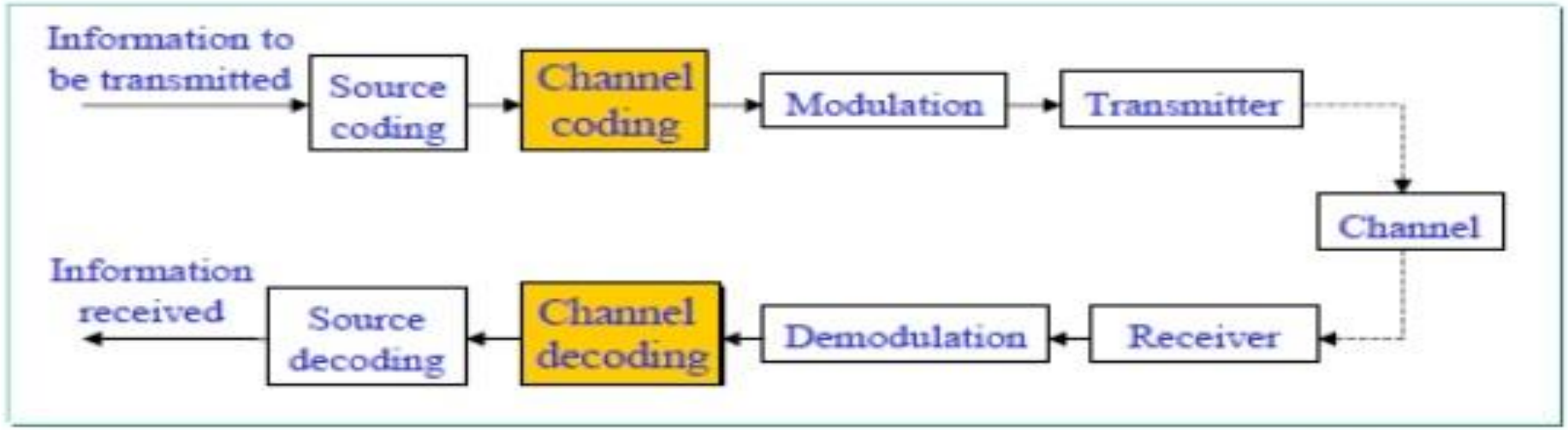
Channel Coding:

Interleaving:

Diversity:


Input Signal
Source Coding
Channel Coding
Modulation
Receive Antenna
Output Signal
Source Decoding
Channel Decoding
Demodulation

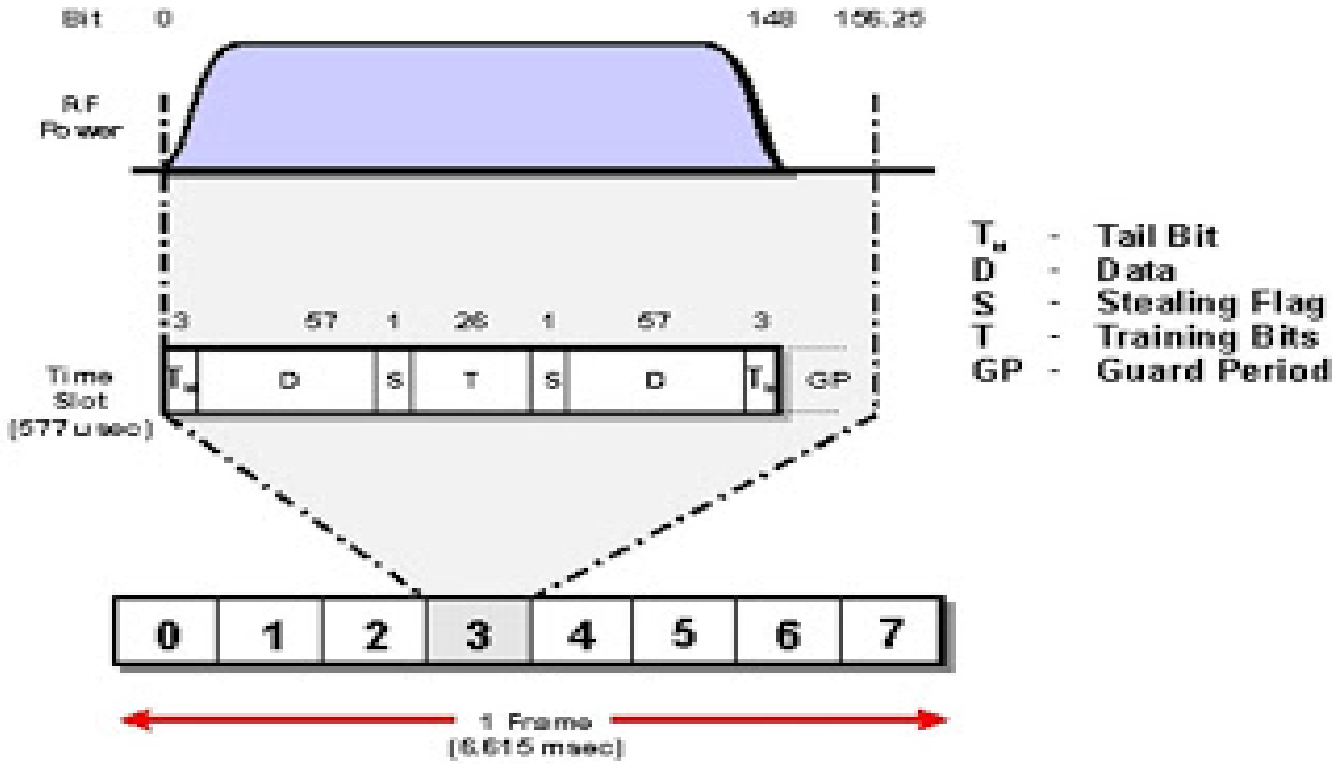
Channel Coding



- Channel coding provides to the receiver the ability of error detection and correction
- Channel Encoder adds some redundant bits to the input bit sequence. Addition of these extra bits
  - Allows the receiver to perform error detection and correction
  - However, increases the bit rate and hence increase required bandwidth

Channel Coding

- **Error Protection**
  - CRC
  - Block
  - Convolutional
- **Signaling Commands**
  - Fast
  - Slow
- **Time Slot Allocation**



Overview of SOURCE encoding, CHANNEL encoding and LINE coding

Source coding

Bandwidth

Redundancy

Channel coding

Error correction & detection

Extra bits

Line coding

0 1 0 1

Pulse shaping



The **interleaving** method is the most effective **coding** method for dispersion of bit errors. The key point of **interleaving** is to disperse some bits(suppose there are “b” bits) of the code into some (suppose to be “n” bursts) burst sequences so as to change the adjacent relationship between bits.

In telecommunications, a **diversity scheme** refers to a method for improving the reliability of a message signal by using two or more communication channels with different characteristics. Diversity is mainly used in radio communication and is a common technique for combatting fading and co-channel interference and avoiding error bursts. It is based on the fact that individual channels experience different levels of fading and interference. Multiple versions of the same signal may be transmitted and/or received and combined in the receiver. Alternatively, a redundant forward error correction code may be added and different parts of the message transmitted over different channels. Diversity techniques may exploit the multipath propagation, resulting in a diversity gain, often measured in decibels.

What are diversity techniques?

Diversity techniques are used in wireless communications systems to primarily to improve performance over a fading radio channel. In such a system, the receiver is provided with multiple copies of the same information signal which are transmitted over two or more real or virtual communication channels

What does antenna diversity mean?

Antenna diversity, also known as space **diversity** or spatial **diversity**, is any one of several wireless **diversity** schemes that uses two or more **antennas** to improve the quality and reliability of a wireless link. ... This is because multiple **antennas** offer a receiver several observations of the same signal

Diversity techniques

The following classes of diversity schemes can be identified:

- **Time diversity**: Multiple versions of the same signal are transmitted at different time instants. Alternatively, a redundant forward error correction code is added and the message is spread in time by means of bit-interleaving before it is transmitted. Thus, error bursts are avoided, which simplifies the error correction.
- **Frequency diversity**: The signal is transmitted using several frequency channels or spread over a wide spectrum that is affected by frequency-selective fading. Later examples include:
  - OFDM modulation in combination with subcarrier interleaving and forward error correction
  - Spread spectrum, for example frequency hopping or DS-CDMA.

•**Space diversity**: The signal is transmitted over several different propagation paths. In the case of wired transmission, this can be achieved by transmitting via multiple wires. In the case of wireless transmission, it can be achieved by antenna diversity using multiple transmitter antennas (transmit diversity) and/or multiple receiving antennas (reception diversity). In the latter case, a diversity combining technique is applied before further signal processing takes place. If the antennas are far apart, for example at different cellular base station sites or WLAN access points, this is called macrodiversity or site diversity. If the antennas are at a distance in the order of one wavelength, this is called microdiversity. A special case is phased antenna arrays, which also can be used for beamforming, MIMO channels and space–time coding (STC).

- **Polarization diversity**: Multiple versions of a signal are transmitted and received via antennas with different polarization. A diversity combining technique is applied on the receiver side.
- **Multiuser diversity**: Multiuser diversity is obtained by opportunistic user scheduling at either the transmitter or the receiver. Opportunistic user scheduling is as follows: at any given time, the transmitter selects the best user among candidate receivers according to the qualities of each channel between the transmitter and each receiver. A receiver must feed back the channel quality information to the transmitter using limited levels of resolution, in order for the transmitter to implement Multiuser diversity.
- **Cooperative diversity**: Achieves antenna diversity gain by using the cooperation of distributed antennas belonging to each node.

Fundamentals of Equalization

Intersymbol interference (ISI)

- caused by multipath propagation (time dispersion) ;
- cause bit errors at the receiver;
- the major obstacle to high speed data transmission over mobile radio channels.

Equalization

- a technique used to combat ISI;
- can be any signal processing operation that minimizes ISI;
- usually track the varying channel adaptively.

Equalization :is the process of remove **ISI** and noise effects from the channel.

It's located at receiver end of the channel. The goal of **equalization** is to mitigate the effects of ISI. However, this goal must be balanced so that in the process of removing **ISI**, the noise power in the received signal is not enhanced

What is frequency hopping used for?

A radio signal whose carrier frequency changes at regular intervals is said to "hop" in frequency from one band to another. Frequency hopping spreads the signal spectrum over a wide band and helps to hide it from interceptors

What is frequency hopping in LTE?

Frequency hopping is one of the techniques that can be used to improve the uplink performance by providing **frequency** diversity and interference averaging. The **hopping** can be between subframes (inter-subframe) or within a subframe (intra-subframe).

Which circuit uses frequency hopping?

The first sub-standard IEEE 802.15. 1 is known as Bluetooth , which uses Frequency Hopping Spread Spectrum (FHSS) . In FHSS the message signal is transmitted by rapidly switching among many frequency channels, using a pseudorandom sequence known to both transmitter and receive